

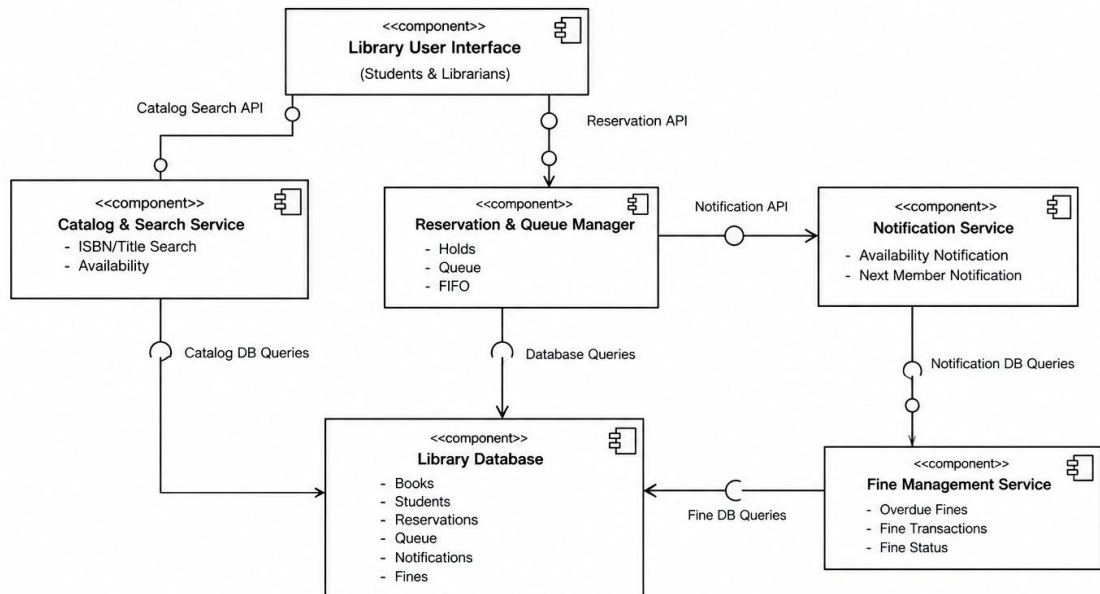
SE LAB 3

SRN: PES2UG24CS315

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TOPIC: DIGITAL CAMPUS LIBRARY RESERVATION GATEWAY

UML COMPONENT DIAGRAM



Chosen Architecture

We chose a Layered Architecture for the Digital Campus Library Reservation Gateway.

Architectural Choice

The system will use a Layered Architecture consisting of presentation, business logic, and data layers. The presentation layer handles student and librarian interactions, the business layer manages catalog searches, reservations, queue processing, notifications and fines, and the data layer stores library information.

Reason 1 – Separation of Concerns

Layered architecture provides clear separation of concerns. Catalog management, reservation and queue processing, notification handling, and fine management can be organized into separate logical components, making the system easier to understand and maintain.

Reason 2 – Scalability and Flexibility

The architecture is suitable for the centralized library system because multiple functions can use the same business logic and database. Changes to the user interface can be made without significantly affecting the underlying data management.

Security Advantage

Security controls can be implemented at the business and data layers so that only authorized users can access student reservations, queue information and fine records. This helps protect sensitive library account information.

Performance Benefit

Frequently used operations such as catalog searches can be handled efficiently by the dedicated catalog service and database layer. Separating responsibilities also prevents unnecessary processing across unrelated components.

Conclusion

Overall, the layered architecture best fits the functional and non-functional requirements of the Digital Campus Library Reservation Gateway.